

Transformations of discrete random variables

▮ You are given the PMF of X .

▮ Let $Y = g(X)$.

▮ The PMF of Y can be obtained as

$$P_Y(y) = P(Y = y)$$

$$= P(g(X) = y)$$

$$= P(X = g^{-1}(y))$$

Ex. Let X have PMF

$$P_X(x) = \begin{cases} \frac{1}{6} & x \in \{1, 2, 3\} \\ 0 & \text{o.w.} \end{cases}$$

Let $Y = g(X) = 3 + X^2$. Find $P_Y(y)$.

$$P_Y(y) = P_X(g^{-1}(y))$$

$$= P_X(\sqrt{y-3}) = \frac{1}{6} \sqrt{y-3}, \quad y \in \{4, 7, 12\}.$$

Let $y = 3 + x^2$, then $x = \sqrt{y-3} = g^{-1}(y)$

Ex. Let X have PMF

$$P_X(x) = \begin{cases} \frac{1}{28} x^2 & , x \in \{\pm 1, \pm 2, \pm 3\} \\ 0 & , \text{o.w.} \end{cases}$$

Let $Y = 3 + X^2$. Find $P_Y(y)$.

$$P_Y(y) = P_X(g^{-1}(y))$$

$$= P(X = -\sqrt{y-3} \text{ or } X = \sqrt{y-3})$$

$$= P_X(-\sqrt{y-3}) + P_X(\sqrt{y-3})$$

$$= 2 \cdot \frac{1}{28} \cdot (y-3), \quad y \in \{4, 7, 12\}.$$

Let $y = 3 + x^2$, then $x = \pm \sqrt{y-3}$.